University of Basel

GOLFIT

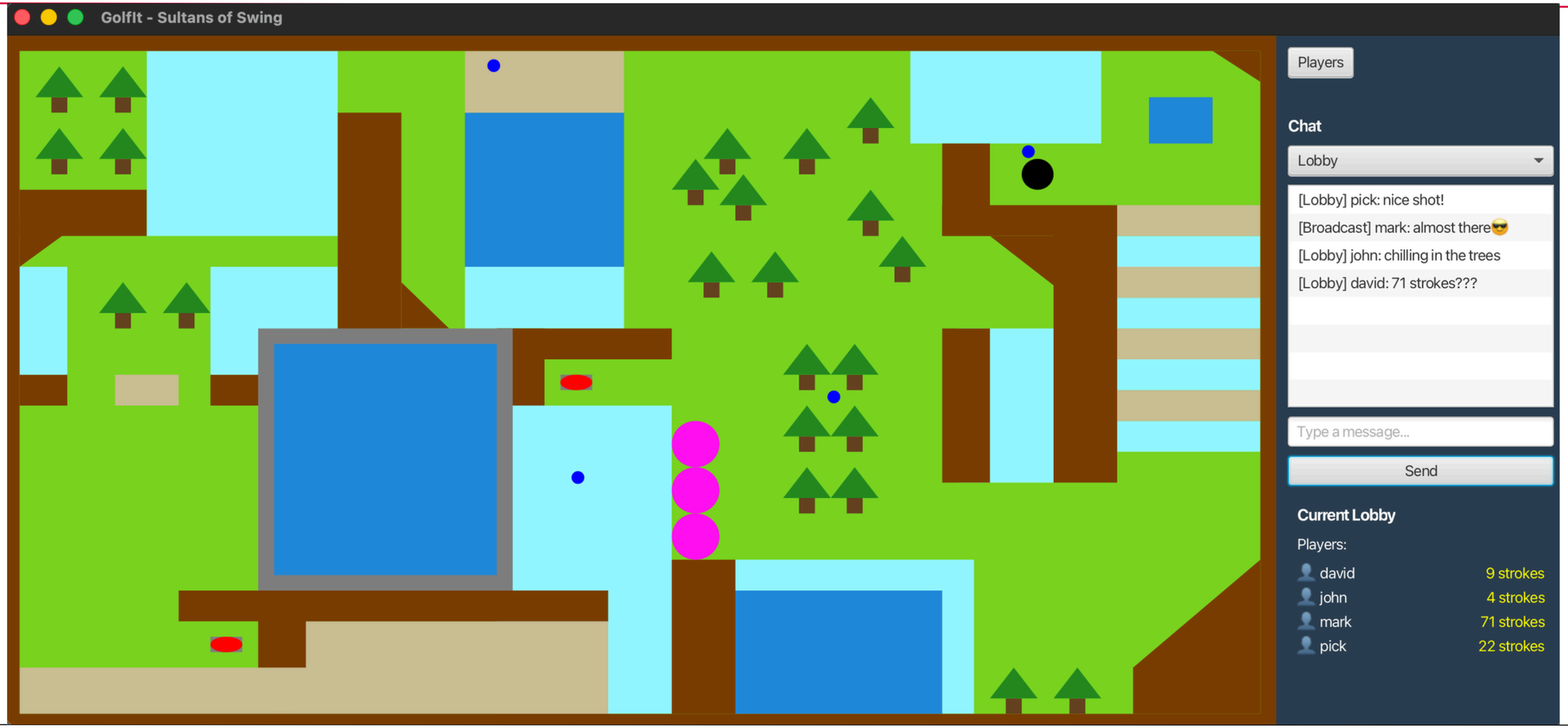
by Sultans of Swing

Tamer, Anton, Demian, Constantin

Rules, Mechanics & Goals:

- Ball→Hole
- Swing and charge the ball
- Bounce to win
- Terrain challenges

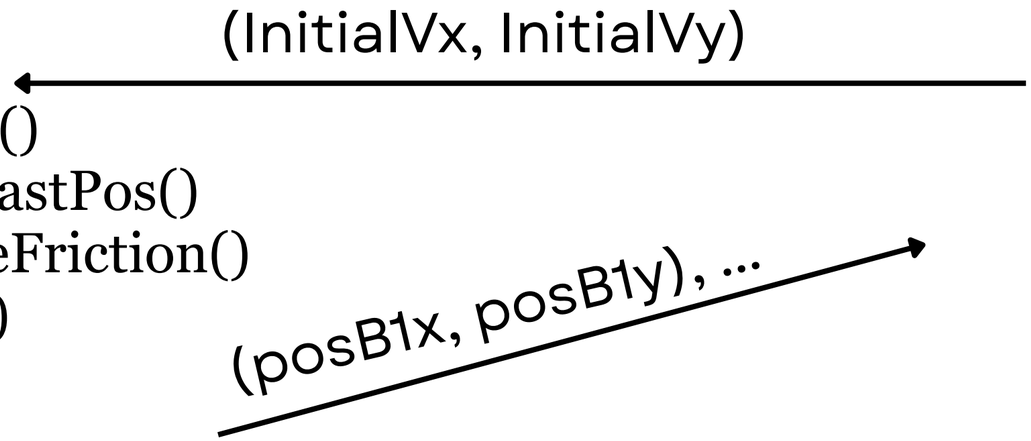
Demo!


GolfIt - Sultans of Swing
Players
Chat
Lobby
[Lobby] pick: nice shot!
[Broadcast] mark: almost there😎
[Lobby] john: chilling in the trees
[Lobby] david: 71 strokes???
Type a message...
Send
Current Lobby
Players:
david 9 strokes
john 4 strokes
mark 71 strokes
pick 22 strokes

Rules to Code

Server

- Logical map:
  - Objects position
  - Terrain Types
- tick 25 times/s
  - checkTerrainType()
    - water->resetLastPos()
    - other->changeFriction()
  - calculateNewPos()
  - resolveCollision()
  - sendBallsPos()



Client

- Visual map with JavaFX
- Send stroke values
- Update ball's nodes

QA Approaches and Technology!

Analytical



Constructive



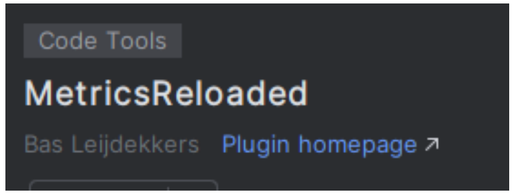
Corrective

Debugging
Regression

QA Metrics

Code quality metrics

Metric	Current value	Target Value	Goal achieved
Unit test coverage	0	>=80%	x
Cyclomatic complexity (avg)	2.19	<=5	✓
Documentation issues	100 (	<=50	x
Static Analysis Issues	59	<=5 (0 high prio)	x



JACOCO
Java Code Coverage

SpotBugs

```
tasks.withType(Javadoc).configureEach { Javadoc it ->
    options.memberLevel = JavadocMemberLevel.PRIVATE
    options.addStringOption('Xdoclint:all', '-quiet')
```

Spotbugs

```java
public class Recursive {
  public void infiniteLoop() {
      infiniteLoop();  // Infinite recursion
  }
}
```




```java
public class Performance {
  public void inefficientMethod() {
      for (int i = 0; i < 1000; i++) {
          String str = new String("Hello");  // Unnecessary object creation
      }
  }
}
```

Progress Report:

- Planning → Protocol Development → Game Logic → GUI
- What’s next ?



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Continuing Education

Thank you!